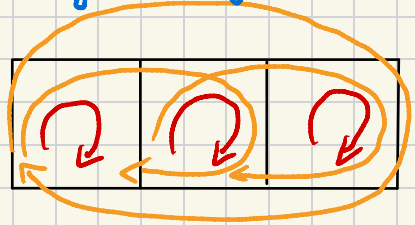
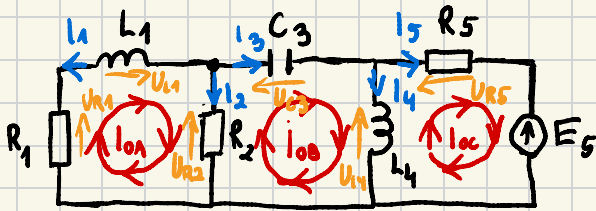


ĆWICZENIA 4

Metoda prądów oczkowych

Mamy różne opcje
wybrania oczek (muszą
obejmować cały układ)



A) $\underline{U}_{R1} + \underline{U}_{L1} + \underline{U}_{R2} = 0$ ← Patrzymy z punktu widzenia I_{0A}

$$\underline{I}_{0A} Z_{R1} + \underline{I}_{0A} Z_{L1} + (\underline{I}_{0A} - \underline{I}_{0B}) Z_{R2} = 0$$

B) $\underline{U}_{R2} + \underline{U}_{C3} + \underline{U}_{L4} = 0$

$$(\underline{I}_{0B} - \underline{I}_{0A}) Z_{R2} + \underline{I}_{0B} Z_{C3} + (\underline{I}_{0B} - \underline{I}_{0C}) Z_{L4} = 0$$

C) $\underline{U}_{L4} + \underline{U}_{R5} + E_5 = 0$

$$(\underline{I}_{0C} - \underline{I}_{0B}) Z_{L4} + \underline{I}_{0C} Z_{R5} = -E_5$$

$$I_1 = -I_{0A}$$

$$I_2 = I_{0A} - I_{0B}$$

$$I_3 = I_{0B}$$

$$I_4 = I_{0B} - I_{0C}$$

$$I_5 = I_{0C}$$

$$\begin{bmatrix} Z_{R1} + Z_{R2} + Z_{L1} & -Z_{R2} & 0 \\ -Z_{R2} & Z_{R2} + Z_{C3} + Z_{L4} & -Z_{L4} \\ 0 & -Z_{L4} & Z_{L4} + Z_{R5} \end{bmatrix} \cdot \begin{bmatrix} I_{0A} \\ I_{0B} \\ I_{0C} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -E_5 \end{bmatrix}$$

Jesli chcemy sprawdzić poprawność prądów to wzywamy NPK!

$$\underline{U}_{R1} = \underline{I}_1 \cdot R_1$$

$$\underline{U}_{C3} = \underline{I}_3 \cdot (-jX_{C3})$$

$$\underline{U}_{L1} = \underline{I}_1 \cdot jX_{L1}$$

$$\underline{U}_{L4} = \underline{I}_4 \cdot jX_{L4}$$

$$\underline{U}_{R2} = \underline{I}_2 \cdot R_2$$

$$\underline{U}_{R5} = \underline{I}_5 \cdot R_5$$

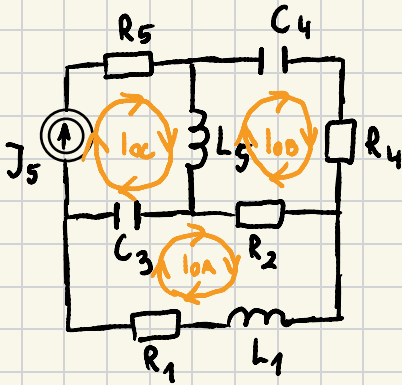
NPK

$$OA : U_{R1} + U_{L1} - U_{R2} = 0$$

$$OB : U_{R2} - U_{C3} - U_{L4} = 0$$

$$OC : U_{L4} - U_{R5} - E_5 = 0$$

Metoda prądów oczkowych,
gdy pojawi się źródło prądowe



I metoda

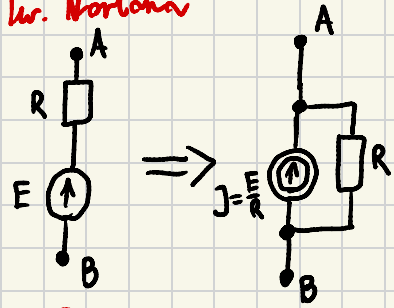
$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & -\underline{Z}_2 & -\underline{Z}_3 \\ -\underline{Z}_2 & \underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5 & -\underline{Z}_5 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_{0A} \\ I_{0B} \\ I_{oc} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ J_5 \end{bmatrix}$$

Piszemy tak dlatego, że
znamy prąd J_5 , ponieważ

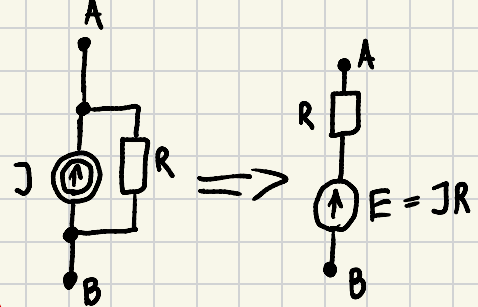
$$I_{oc} = J_5$$

$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 & -\underline{Z}_2 \\ -\underline{Z}_2 & \underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5 \end{bmatrix} \cdot \begin{bmatrix} I_{0A} \\ I_{0B} \end{bmatrix} = \begin{bmatrix} \underline{Z}_3 J_5 \\ \underline{Z}_5 J_5 \end{bmatrix}$$

Tw. Nortona



Tw. Thevenina



II metoda

$$I_{0A}(\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3) - I_{0B}\underline{Z}_2 - I_{0C}\underline{Z}_3 = 0$$

$$-I_{0A}\underline{Z}_2 + I_{0B}(\underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5) - I_{0C}\underline{Z}_5 = 0$$

$$I_{0C} = J_5$$

$$\begin{cases} I_{0A}(\underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3) - I_{0B}\underline{Z}_2 = \underline{Z}_3 J_5 \\ -I_{0A}\underline{Z}_2 + I_{0B}(\underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5) = \underline{Z}_5 J_5 \end{cases}$$

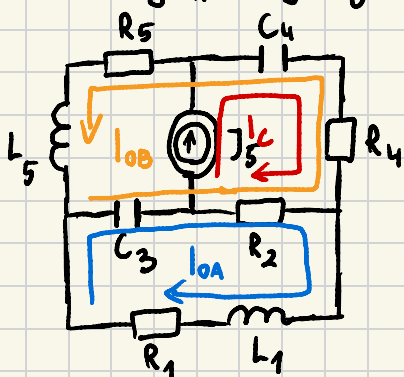
$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 & -\underline{Z}_2 \\ -\underline{Z}_2 & \underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5 \end{bmatrix} \cdot \begin{bmatrix} I_{0A} \\ I_{0B} \end{bmatrix} = \begin{bmatrix} \underline{Z}_3 J_5 \\ \underline{Z}_5 J_5 \end{bmatrix}$$

Pamiętać o dobrej kolejności,
aby nie zrobić złe macierzy

I_{0A} I_{0B} I_{0C}

Metoda prądów oczkowych, gdy źródło prądowe

poędzy "domyślnymi" oczkami

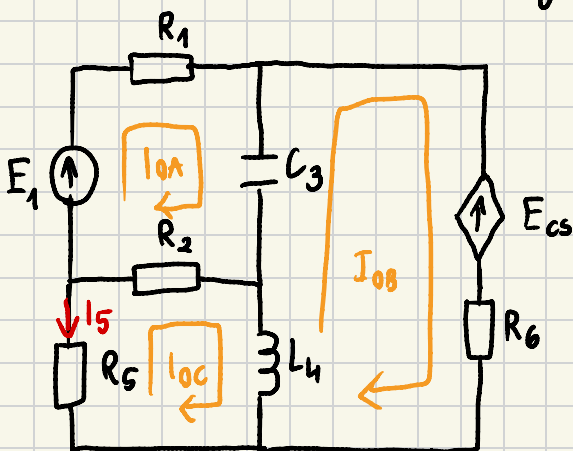


Robimy oczka, tak by tylko jedno oczko zawierało źródło prądowe

$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & \underline{Z}_2 + \underline{Z}_3 & -\underline{Z}_2 \\ \underline{Z}_2 + \underline{Z}_3 & \underline{Z}_2 + \underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5 & -\underline{Z}_4 - \underline{Z}_2 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} I_{OA} \\ I_{OB} \\ I_{OC} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ J_5 \end{bmatrix}$$

$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & \underline{Z}_2 + \underline{Z}_3 \\ \underline{Z}_2 + \underline{Z}_3 & \underline{Z}_2 + \underline{Z}_3 + \underline{Z}_4 + \underline{Z}_5 \end{bmatrix} \cdot \begin{bmatrix} I_{OA} \\ I_{OB} \end{bmatrix} = \begin{bmatrix} \underline{Z}_2 J_5 \\ (\underline{Z}_2 + \underline{Z}_4) J_5 \end{bmatrix}$$

Metoda prądowa oczkowa, gdy mamy źródło sterowane napięciem



$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & -\underline{Z}_3 & -\underline{Z}_2 \\ -\underline{Z}_3 & \underline{Z}_3 + \underline{Z}_4 + \underline{Z}_6 & -\underline{Z}_4 \\ -\underline{Z}_2 & -\underline{Z}_4 & \underline{Z}_4 + \underline{Z}_5 + \underline{Z}_2 \end{bmatrix} \cdot \begin{bmatrix} I_{0A} \\ I_{0B} \\ I_{0C} \end{bmatrix} = \begin{bmatrix} E_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -E_{cs} \\ 0 \end{bmatrix}$$

Wyznaczamy E_{cs} i podkładamy do równania

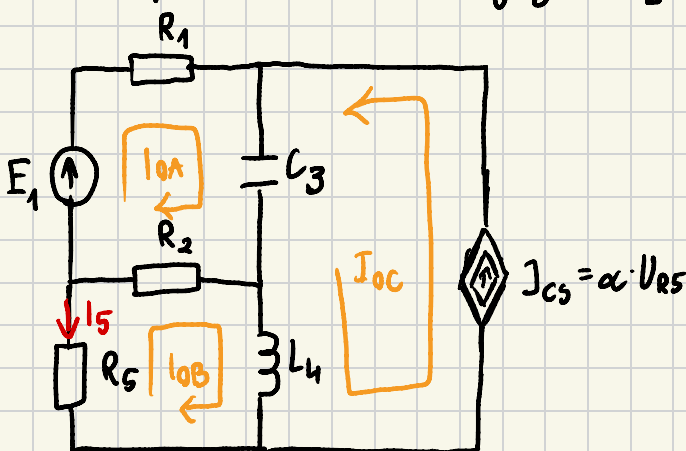
$$I_5 = -I_{0C}$$

$$E_{cs} = \alpha \cdot U_{R5} = \alpha \cdot R_5 \cdot (-I_{0C})$$

$$\begin{bmatrix} 0 \\ \alpha R_5 I_{0C} \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & -\underline{Z}_3 & -\underline{Z}_2 \\ -\underline{Z}_3 & \underline{Z}_3 + \underline{Z}_4 + \underline{Z}_6 & -\underline{Z}_4 - \alpha R_5 \\ -\underline{Z}_2 & -\underline{Z}_4 & \underline{Z}_4 + \underline{Z}_5 + \underline{Z}_2 \end{bmatrix} \cdot \begin{bmatrix} I_{0A} \\ I_{0B} \\ I_{0C} \end{bmatrix} = \begin{bmatrix} E_1 \\ 0 \\ 0 \end{bmatrix}$$

Metoda prądowo oczkowa, gdy mamy źródło sterowane prądowe



$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & -\underline{Z}_2 & \underline{Z}_3 \\ -\underline{Z}_2 & \underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5 & \underline{Z}_4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} E_1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ J_{cs} \end{bmatrix}$$

Wyznaczymy J_{cs} i podstawimy do równania

$$J_{cs} = \alpha \cdot U_{RS} = \alpha \cdot R_5 \cdot I_5 = -\alpha R_5 \cdot I_{ob}$$

$$\begin{bmatrix} \underline{Z}_1 + \underline{Z}_2 + \underline{Z}_3 & -\underline{Z}_2 & \underline{Z}_3 \\ -\underline{Z}_2 & \underline{Z}_2 + \underline{Z}_4 + \underline{Z}_5 & \underline{Z}_4 \\ 0 & \alpha R_5 & 1 \end{bmatrix} = \begin{bmatrix} E_1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ -\alpha R_5 I_{ob} \end{bmatrix}$$